CAT-I (D1+TD1) (24BCE2017)

(Tharsan kirthic.K)

Q1)

a) i) error   correct : MUL AB

→ Comma not be used

ii) error   correct: push Acc

→ accepts only direct address

iii) error   Correct: ANL A, R_0

→ destination must be accumallator or direct address

iv) error

→ $R_2 - R_7$ cannot be used as pointers.

→ only R_0 & R_1 supports indirect addressing

v) error

→ Registers to register moves not allowed.

b) i) Direct addressing mode

ii) Immediate addressing mode

iii) Register addressing mode

iv) Indexed addressing mode

v) Register Indirect Addressing mode

(Thassankirthi.C.A) (24BCB2017)

2) ORG 0000H

```
LJMP MAIN
ORG 0030H

MAIN:
    MOV DPTR, # 250H;
    MOV R0 , # #40H;
    MOV R2 ,#06H;

LOAD_DATA :
    CLR A;
    MOVC A ,@A+DPTR;
    MOV @R0 ,A;
    INC DPTR;
    INC R0;
    DJNZ R2 ,LOAD_DATA ;
    MOV R0 ,#40H;
    MOV R2 ,# 05H;
    MOV A ,@R0 ;
    MOV R4, #00H;

ADD_LOOP:
    INC R0;
    ADD A ,@R0 ;
    DA A;
    JNC NEXT;
    INC R4;
```

(24BCE2017) (Tharan Kirthic A)

NEXT:

```
DJNZ  RL , ADD_LOOP;
MOV   R5 ,A;
H: SJMP H;

org  250 H
DATA_1: DB . 54H, 76H, 65H, 98H, 88H, 99H
END
```

;Higher byte in R3
;lower byte in R4

~~3)~~

```
~~org   0000H~~
~~LJMP  MAIN~~
~~org~~
```

3)

```
ORG     0000H
MAIN:
JB . P1.0 , HIGH
LOW:
MOV    A,P0
MOV    P3 ,A
SETB   P2.1
ACALL  DELAY
CLR P2.1
SJMP  MAIN
```

(24BCE2017)(Thanvankarthic.K)

```
AIGH:

MOV    A,P0

MOV    P2,A

SETB   P3.1

ACALL  DELAY

CLR    P3.1

SJMP   MAIN

DELAY:

MOV    R5,#255

D1  : MOV    R6,#255

D2  : DJNZ   R6,D2

DJNZ   R5,D1

RET

END
```

(Thasonkirthica) (24BCE2017)

4) Ans:

	Stack Content	SP
push 0	08H: 30H	08H
push 1	09H : 23H	09H
push 2	0AH : 00H	0AH
push 3	0BH : 24H	0BH
pop 3	24H removed	0AH
pop 7	00H removed	09H
push 3	0AH: 24H	0AH
pop 6	24H removed	09H

Reg / location	value in Hex
	30H
R_0	24H
R_1	24H
R_3	00H
R_7	24H
R_6	

(24BCE2017) (Thorsan Kirthi.K)

5) Ans:

(i) Ans:

(a) "78H" is stored in 12H (R2)

" A4H" is stored in 14H (R4)

Since Bank 3 is selected ($RS_1 = 1$ & $R_0 = 1$)

SET B PSW.4 ⇒ Sets bit 4 of PSW [$RS_1 = 1$]

MOV R2, #78H.

MOV A, R2 ⇒ Sets bit 3 of PSW ($RS0 = 1$)

Set B PSW.3

MOV R4, #0A4H

ADD A, R4

END

b) After the execution of instruction.

"ADD A, R4";

status of flags;

Carry (cy) = 1

Auxilary Carry (AC) = 0

overflow (ov) = 1

(24BCB2019) (Tharun Kirthic.M)

ii)

vallu in Accumlator after each instruction

Instruction	A (Hex value)
ANL A,#00H	00H
MOV A,3AH	3AH
SWAP A	A3H
ANL A,# 23H	23H
OR AB	08H

CAT-II (D2+TD2) Thosson Krishnak
(24BCE2017) (24-25)

Q) i) (a) Ans :

MOV A, D0.1 (wrong)

correct Instruction : mov C, D0.1

ii) DEC DPTR (wrong)

⇒ It is not possible to decrement DPTR register

we can only increment it.

⇒ Correct Instruction : INC DPTR

iii) error

can't increment on immediate value

Correct : INC 40H

iv) SETB A is not correct

↳ SetB is bit-addressable but accumulator is not bit-addressable in this way.

Correct : mov A, # 04OH

v) error

↳ There is no 'B' after CLR

Correct: CLR ~~B~~ $P_2.0$

(Thasson Arithic.K) (24BCB2017)

b) $T = \frac{12}{f} = \frac{12}{15 \times 10^6 \, Mz} = 0.8\,\mu s$.

instruction	miching cycle
MOV R3, #50H	1
HERE : NOP	1
NOP	1
NOP	1
NOP	1
~~LOP~~ DJNZ R3, HERE	2
RET	2

total cycles :-

→ instruction : 1 cycle

→ loop body : $(1+1+1+2) \times 80 = 6 \times 80 = 480$

→ Exit (RET) = 2 cycles

→ Total : $1+480+2 = 483$ cycle

total time

$= 483 \times 0.8\,\mu s$

$= 386.4\,\mu s$

(24BEE2017) (Tharsan Karthic K)

2)
```
        org   0000H

START : MOV    A , P1;

        JB  ACC.7 , SQUARE_25;

wav50 : SETB  P2.1;

        ACALL DELAY_3
        CLR  P2.1;
        ACALL DELAY_2
        SJMP  START

SQUARE_25 :
        SETB  P2.1;
        ACALL  DELAY_1;
        CLR  P2.1;
        ACALL  DELAY_3;
        SJMP   START

DELAY_3 : MOV   R0 , #30;

D3  : DJNZ   R0 , D3;
        RET;

DELAY_2:   MOV R0,#20;

D2 :   DJNZ   R0, D2;

        RET;

DELAY_1:   MOV  R0 , #10

D1:    DJNZ  R0, D1;
        RET;

END
```

(24BCE2017) (Tharsan Kirthic. K)

3)
```
org 0000H
MOV DPTR ,#300H
MOV R1 ,#10
MOV R2,# 00H
MOV R3 ,# 00H;

BACK : CLR A;
       MOVC A,@A+DPTR ,
       ADD A,R2 ;
       MOV R2 ,A;
       JNC NEXT;
       INC R3;

NEXT : INC DPTR;
       DJNZ R1 ,BACK :

HERE:  SJMP HERE ;
       org 300H ;
       DB 92H ,34H ,84H , 29H ,22H ,11H ,33H,34H,
          55H,66H;

END.
```

4)

(24BCB20A)(Tharsan kirthic K)

	data in RAM	SP
MOV 05H, #3EH;	05H .3BH	07H
MOV 0AH, #DFH;	0AH : DFH	07H
MOV 2FH, #45H;	2FH : 45H	07H
MOV SP, #69H;		69H
push 05H;	6AH : 3B H	6AH
push 0AH;	6BH : DFH	6BH
push 2FH;	6CH : 45H	6CH
pop 4AH;	4AH : 45 H	6BH
pop 4BH;	4BH : DFH	6AH
push 4BH;	6BH : DFH	6BH
pop 5AH;	5AH : DFH	6AH

(24BCE2081) (Tharson Kinthick)

5)

i)

Instruction	Accumulator	C	RAM
			–
MOV A, #83H	83H (10000011)	0	
RLC A	06H (00000110)	1	–
RLC A	0DH (00001101)	0	–
MOV 20H, #87H	0DH	0	87H
ANL C, D2 H	0DH	0	87H

~~Final Result:-~~

Final Result:

A: 0DH

RAM [20H] : 87H

Carry Flag : 0

(ii)

C24BCB2017 - Tharsan Karthick

ii)

Instruction	Register A	Register R0	RAM [40H]	C
MOV A,#80H	80H	–		0
MOV R0,#40H	80H	40H	–	0
MOV @R0,A	80H	40H	–	0
INC A	81H	40H	80H	0
RRC A	40H	40H	80H	0
ADD A,40H	C0H	40H	80H	1
			80H	0

Final Result:

A: C0H

R0: 40H

RAM [40H] : 80H

Carry Flag : 0